

Physics 4730, Fall 2025: Lab 09

Please complete the calculations below in class. We will make use of the results in Lab 10.

1. The discrete Poisson probability density function is given by:

$$f(x|\lambda) = \frac{\lambda^x}{x!} e^{-\lambda}$$

where x is a nonnegative integer. Write down an expression for the *likelihood function* L for a set of measurements x_i of a Poisson random variable. Apply any simplifications you can think of.

$$\begin{aligned} L &= \prod_{i=1}^N \frac{\lambda^{x_i}}{x_i!} e^{-\lambda} \\ &= e^{-N\lambda} \prod_{i=1}^N \frac{\lambda^{x_i}}{x_i!} \end{aligned}$$

2. Now, find the *log-likelihood* function $\ln(L)$. Apply identities to break it into three terms: one dependent on x_i , one dependent on $x_i!$, and a constant.

$$\begin{aligned} \ell = \ln(L) &= -N\lambda + \sum_{i=1}^N \ln\left(\frac{\lambda^{x_i}}{x_i!}\right) \\ &= -N\lambda - \sum_{i=1}^N \ln(x_i!) + \ln\lambda \sum_{i=1}^N x_i \end{aligned}$$

3. Finally, by maximizing $\ln(L)$ with respect to λ , find an expression for the MLE for λ .

$$\left. \frac{\partial \ell}{\partial \lambda} \right|_{\lambda=\lambda_0} = -N + \frac{1}{\lambda_0} \sum_{i=1}^N x_i = 0$$

$$\boxed{\lambda_0 = \frac{1}{N} \sum_{i=1}^N x_i}$$